

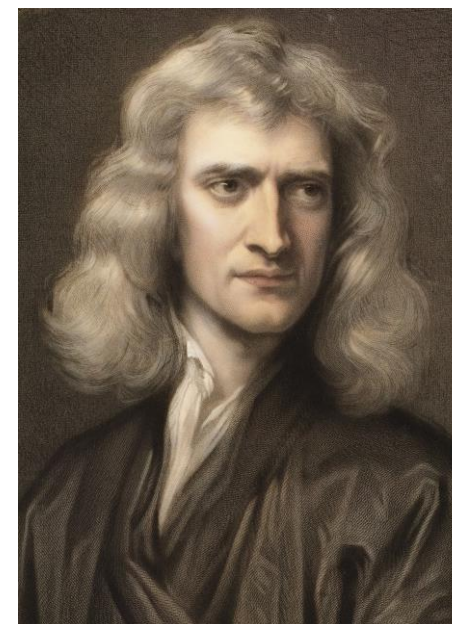
Math Primer

(6. Integration)

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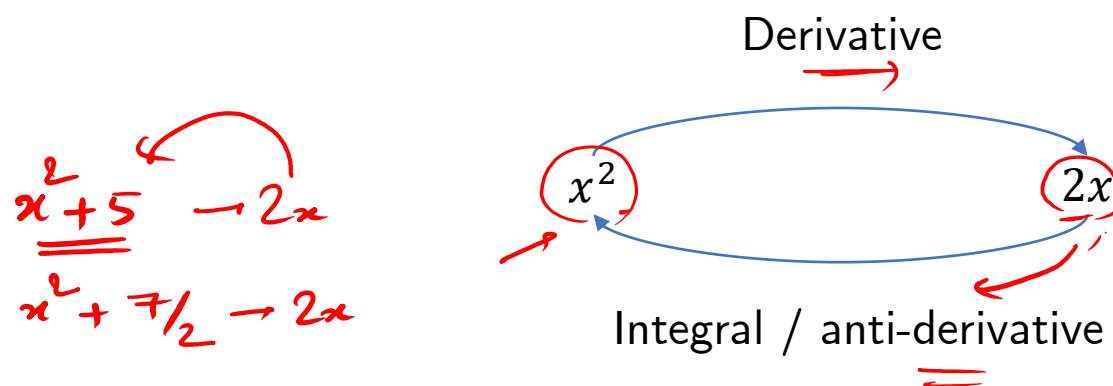
Isaac Newton
(1643-1727)



G.W. Leibniz
(1646-1716)

Integrals as anti-derivatives

- ▶ Derivative of a function $f(x)$ gives the slope of the function $f'(x)$. If x is a free input variable, the derivative itself is a net new function. Example, $f(x) = x^2$, then $f'(x) = 2x$ which is another function
- ▶ Naturally, the next question is given the derivative of a function, what is the function? It's a reverse engineering problem and harder to calculate than derivatives.



- ▶ We will shortly see that the anti-derivative is **not unique**

Properties of Indefinite Integral

▶ When we arrive at the integral in a functional form, it's called indefinite integral.

▶ The notation for indefinite integral is

$$\int (\cdot) dx$$

$$\int (2x) dx = x^2 + C$$

▶ The function that is being integrated is called Integrand and x is the integration variable

▶ Because the derivative of a constant is 0, we add it to the known anti-derivative to express the most general function possible.

▶ Therefore

$$\int f'(x) dx = f(x) + C$$

C is an arbitrary constant (sometimes called as constant of integration)

Properties of Indefinite Integral

Some basic Integration formulae

Handwritten notes for integration of logarithmic functions:

- $x > 0, \ln(x) \rightarrow \frac{1}{x}$
- $x < 0, \ln(-x) \rightarrow \frac{1}{-x} \times (-1) = \frac{1}{x}$
- $\ln|x| \rightarrow \frac{1}{x}$

$$\int x^n dx = \frac{x^{n+1}}{n+1} + C$$

$$\int 1 dx = x + C$$

$$\int e^x dx = e^x + C$$

$$\int \frac{1}{x} dx = \ln|x| + C$$

Handwritten derivation of the power rule:

$$\boxed{\frac{x^{n+1}}{n+1}} \rightarrow \frac{1}{n+1} \cdot (n+1)x^n \rightarrow x^n$$

Properties of Indefinite Integral

► Integral is a Linear operator $\rightarrow \left. \begin{aligned} \text{op}(x \pm y) &= \text{op}(x) \pm \text{op}(y) \\ \text{op}(\alpha x) &= \alpha \text{op}(x) \end{aligned} \right\}$

$$\int (\alpha \overset{\downarrow}{f'(x)} \pm \beta \overset{\downarrow}{g'(x)}) dx = \alpha \overset{\uparrow}{f(x)} \pm \beta \overset{\uparrow}{g(x)} + \textcircled{C}$$

Exercise 1: $\int (3x^2 + 4x^3) dx \rightarrow 3 \int x^2 dx + 4 \int x^3 dx = 3 \cdot \frac{x^3}{3} + 4 \cdot \frac{x^4}{4} + C = x^3 + x^4 + C$

Exercise 2: $\int \left(5e^x - \frac{4}{x} \right) dx \rightarrow 5 \int e^x dx - 4 \int \frac{1}{x} dx = 5e^x - 4 \ln|x| + C$
 $-x^{-2} \rightarrow -\frac{x^{-2+1}}{-2+1} = \frac{1}{x}$

Exercise 3: $\int \left(\frac{x^3 - 1}{x^2} \right) dx \rightarrow \int \left(x - \frac{1}{x^2} \right) dx = \frac{x^2}{2} + \frac{1}{x} + C$

Methods of Integration – Substitution

► Most of the times, the integrand will be a composite function of x . In this case, we make use of intermediate variables (method of **substitution**) which is the reverse of chain rule.

► So, if we have $\int f(g(x))dx$ then assume $g(x) = t$, so $\frac{dt}{dx} = g'(x)$

$$y = e^{5x} \rightarrow \begin{aligned} t &= 5x \\ \frac{dt}{dx} &= 5 \\ \frac{dy}{dt} \cdot \frac{dt}{dx} &= e^t \times 5 \\ &= e^{5x} \cdot 5 \end{aligned}$$

Exercise 1 $\int e^{5x} dx \rightarrow \int e^t \frac{dt}{5} = \frac{1}{5} e^t + C = \frac{e^{5x}}{5} + C$

Exercise 2 $\int (2x \cos(x^2 + 1)) dx \rightarrow \int \cos(t) dt = \sin t + C = \sin(x^2 + 1) + C$
 $\begin{aligned} x^2 + 1 &= t \\ 2x &= \frac{dt}{dx} \end{aligned}$

Exercise 3 $\int \frac{2x}{1+x^2} dx \rightarrow \int \frac{dt}{t} = \ln|t| + C = \ln|1+x^2| + C$

Exercise 4 $\int \frac{1}{x + x \ln x} dx \rightarrow \int \frac{1}{x(1 + \ln x)} dx = \int \frac{dt}{t} = \ln|t| + C = \ln|1 + \ln x| + C$
 $\begin{aligned} t &= 1 + \ln x \\ \frac{dt}{dx} &= \frac{1}{x} \end{aligned}$

Exercise 5 $\int \frac{e^{2x} - 1}{e^{2x} + 1} dx \rightarrow \int \frac{e^x - e^{-x}}{e^x + e^{-x}} dx = \int \frac{dt}{t} = \ln|e^x + e^{-x}| + C$
 $\begin{aligned} t &= e^x + e^{-x} \\ \frac{dt}{dx} &= e^x - e^{-x} \end{aligned}$

Methods of Integration – Substitution

Extra references

$$\begin{aligned}
 \int \frac{dx}{x^2 - a^2} &= \frac{1}{2a} \ln \left| \frac{x-a}{x+a} \right| + C \quad \rightarrow \quad \frac{1}{x^2 - a^2} = \frac{1}{(x+a)(x-a)} \equiv \frac{1}{2a} \frac{(x+a) - (x-a)}{(x+a)(x-a)} = \frac{1}{2a} \left(\frac{1}{x-a} - \frac{1}{x+a} \right) \\
 &\quad \Rightarrow \int (\cdot) dx = \frac{1}{2a} \left(\ln|x-a| - \ln|x+a| \right) + C \\
 &\quad = \frac{1}{2a} \ln \left| \frac{x-a}{x+a} \right| + C
 \end{aligned}$$

$$\int \frac{dx}{x^2 + a^2} = \frac{1}{a} \tan^{-1} \left(\frac{x}{a} \right) + C$$

$$\int \frac{dx}{\sqrt{x^2 - a^2}} = \ln \left| x + \sqrt{x^2 - a^2} \right| + C$$

$$\int \frac{dx}{\sqrt{a^2 - x^2}} = \sin^{-1} \left(\frac{x}{a} \right) + C$$

Methods of Integration – Partial Fractions

- Below are some specific handling of integrands when they are rational functions. Equate coefficients on RHS with LHS for powers of x . RHS is easy to integrate.

Rational Function	Partial Fractions
$\frac{px + q}{(x - a)(x - b)}$	$= \frac{A}{x - a} + \frac{B}{x - b}$
$\frac{px + q}{(x - a)^2}$	$\frac{A}{x - a} + \frac{B}{(x - a)^2}$
$\frac{px^2 + qx + r}{(x - a)(x - b)(x - c)}$	$\frac{A}{x - a} + \frac{B}{x - b} + \frac{C}{x - c}$
$\frac{px^2 + qx + r}{(x - a)^2(x - b)}$	$\frac{A}{x - a} + \frac{B}{(x - a)^2} + \frac{C}{x - b}$
$\frac{px^2 + qx + r}{(x - a)(x^2 + bx + c)}$	$\frac{A}{x - a} + \frac{Bx + C}{x^2 + bx + c}$

Methods of Integration – Partial Fractions



Exercises – integrate the following

$$\int \frac{1}{(x+1)(x+2)} dx \quad \rightarrow$$

$$\int \frac{x}{(x-1)^2} dx$$

$$\int \frac{1}{x(x^2+1)} dx$$

$$\int \frac{1}{x^4-1} dx$$

Methods of Integration – By parts

- Integration of product of two functions is given by **integration by parts** formula. It's the reverse of product rule of derivative

$$\int \ln x \, dx \quad \int \frac{1}{x} \, dx$$

$$\int \underline{(uv)} \, dx = \underline{u} \int \underline{v \, dx} - \int \left[\underline{u'} \int \underline{v \, dx} \right] \, dx$$

How to choose u and v ? Well, there is a rule of thumb for u

It's called **ILATE** (Inverse Trig, Log , Algebraic, Trig, Exponential)

Exercise 1 $\int x \cos x \, dx$ →

Exercise 2 $\int \ln x \, dx$

Exercise 3 $\int x e^x \, dx$

Exercise 4 $\int e^x \sin x \, dx$

Definite Integral

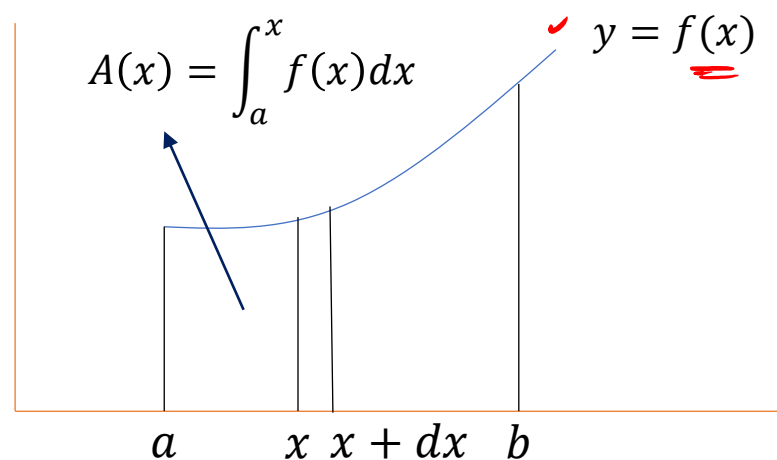


- ▶ In contrast to indefinite integral which is a function, a definite integral is a unique value. It's denoted as

$$\int_a^b f(x) dx$$

a is the lower limit and b is the upper limit

- ▶ Graphically, it's the area under the curve $f(x)$ from a to b
- ▶ The value of this integral is given as $F(b) - F(a)$, where $F(x)$ is the antiderivative of $f(x)$



Definite Integral

Exercises – integrate the following

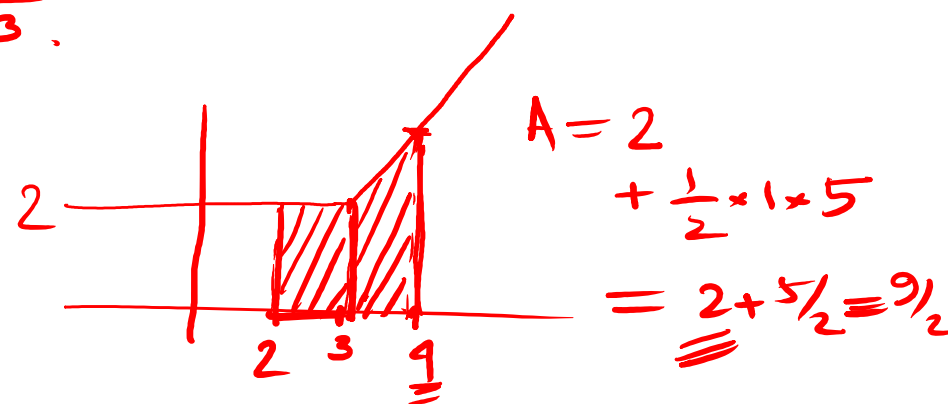
$$\int_{-2}^2 x dx = \left[\frac{x^2}{2} \right]_{-2}^2 = 0$$

$$\int_{-a}^a f(x) dx = 0 \text{ if } f(x) = \text{odd function.}$$

$$2 \int_0^a f(x) dx = \text{even function.}$$

$$\int_{-2}^2 x^2 dx = \left[\frac{x^3}{3} \right]_{-2}^2 = \frac{8}{3} + \frac{8}{3} = \frac{16}{3}$$

$$\int_2^4 f(x) dx, \text{ where } f(x) = \begin{cases} 2, & x < 3 \\ x-1, & x \geq 3 \end{cases}$$



$$= \int_2^3 2 dx + \int_3^4 (x-1) dx$$

$$= \left[2x \right]_2^3 + \left[\frac{x^2}{2} - x \right]_3^4 = 2 + \left(4 - \left(\frac{9}{2} - 3 \right) \right) = 2 + \frac{5}{2} = \frac{9}{2}$$

Definite Integral – With substitution

- When we perform variable transformation as part of substitution, we also need to change the limits based on the new variable used

$$\int_0^1 \frac{x}{x^2 + 1} dx \rightarrow \frac{1}{2} \int \frac{dt}{t} = \frac{1}{2} \ln|t| = \frac{1}{2} \ln(x^2 + 1), \quad \left[\frac{1}{2} \ln(x^2 + 1) \right]_0^1 = \frac{\ln 2}{2} \checkmark$$

$$\left[\frac{1}{2} \ln(t) \right]_1^2 = \frac{1}{2} \ln 2 \checkmark$$

$$\int_0^2 x \sqrt{x+2} dx, \quad (x+2) = t^2$$

$$\Rightarrow 1 = 2t \cdot dt/dx \Rightarrow dx = 2t \cdot dt$$

$$= \int_{\sqrt{2}}^2 (t^2 - 2) \cdot t \cdot 2t \cdot dt = \int_{\sqrt{2}}^2 (2t^4 - 4t^2) dt$$

Definite Integral – Properties

Some very useful formulas to calculate definite integrals

$$\int_a^b f(x) dx = \int_a^b f(t) dt$$

$$\int_a^b f(x) dx = F(b) - F(a) = -(F(a) - F(b)) = -\int_b^a f(x) dx$$

$$\int_a^b f(x) dx = -\int_b^a f(x) dx$$

$$\int_a^a f(x) dx = 0$$

$$\int_a^a f(x) dx = -\int_a^a f(x) dx$$

$$\int_a^b f(x) dx = \int_a^b f(a+b-x) dx$$

$$a+b-x=t, -dx=dt$$

$$-\int_b^a f(t) dt = \int_a^b f(t) dt$$

$$\int_0^a f(x) dx = \int_0^a f(a-x) dx$$

$$a-x=t, -dx=dt$$

$$-\int_a^0 f(t) dt = \int_0^a f(t) dt$$

$$\int_{-a}^a f(x) dx = 2 \int_0^a f(x) dx, \text{ If } f(x) \text{ is even, else } 0$$